CODECHELLA MADRID 2026

Staggered DiD: Bacon + Callaway-Sant'Anna

Day 4: Staggered DiD

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Stevenson & Wolfers (2006): can law reduce intimate violence?

Before 1969, divorce in the U.S. required showing fault — adultery, abandonment, cruelty — and either spouse could veto. Starting with California in 1969, states began adopting **unilateral divorce** laws: either spouse can dissolve the marriage, no fault required, no veto. By 1985 most states had moved.

The theory. Lower exit cost shifts bargaining power within marriage. A woman with a credible threat to leave faces less violence and depression. Less violence, less depression $\Rightarrow$ fewer suicides.

The question. Did the staggered timing of unilateral-divorce reforms reduce women's suicide rates?

Stevenson, B., and J. Wolfers (2006). "Bargaining in the Shadow of the Law: Divorce Laws and Family Distress." QJE 121(1): 267–288.

Their data and design

Data. NCHS Multiple Cause of Death (1964–1996); 1960 Census $+$ SEER for population denominators; SW's own coding of state divorce-law dates.

Outcome. Age-adjusted female suicide rate per million women.

Sample. 37 states that adopted unilateral divorce 1969–1985 (staggered); 14 control states (5 never-reform, 8 pre-1964 reform, 1 excluded).

The design. Staggered two-way fixed effects:

$$y_{ist} = \alpha_i + \gamma_t + \delta D_{ist} + \varepsilon_{ist}$$

Their headline. An 8–16% decline in women's suicide rate in unilateral-divorce states, growing over time. Companion declines in domestic violence and intimate-partner homicide.

Their event study: the effect grows over time

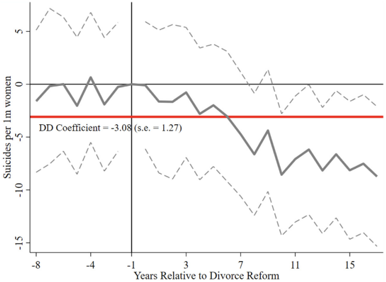
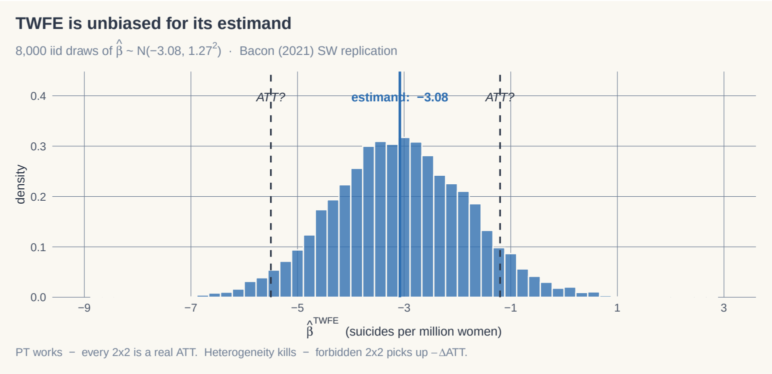


Fig. 5. Event-study and difference-in-differences estimates of the effect of no-fault divorce on female suicide: Replication of Stevenson and Wolfers (2006). Notes: The figure plots event-study estimates from the two-way fixed effects regression equation on page 276 and plotted in Fig. 1 of Stevenson and Wolfers (2006), along with the DD coefficient. The specification does not include other controls and does not weight by population. Standard errors are robust to heteroskedasticity.

*Event study of no-fault divorce on female suicide (replication of SW from Bacon 2021, Fig. 5). Post-reform coefficients drift down to ≈ -10 per million women. Red line: static TWFE returns -3.08 **(s.e. 1.27)** — much smaller than the event-study average.*

Their finding -3.08 (s.e. 1.27). Is it unbiased?



PT works (every 2×2 is a real ATT) · **Heterogeneity kills** (forbidden 2×2 picks up $-\Delta\text{ATT}_k$)

What *is* the estimand? Bacon decomposes it into 156 components

Bacon (2021) writes the TWFE estimand as a weighted average of underlying 2×2 DiDs. On SW data: 156 components.

- Late-treated-vs-already-treated 2×2s average $+3.51$ — the *wrong sign*, pulled there by treatment-effect heterogeneity over time.
- Drop those terms: the rest averages -5.44, very close to the event-study average of -4.92 from the figure two slides back.

So the estimand -3.08 is averaging $+3.51$ with -5.44. Both pieces come from the same regression on the same data. **Whether either piece is the ATT depends on what we assume.**

That's what this morning develops. *Goodman-Bacon, A. (2021), J. Econometrics 225(2): 254–277. Decomposition in Fig. 6.*

This morning, two acts

1. **Why TWFE goes wrong (Goodman-Bacon 2021).** TWFE is a variance-weighted average of every 2$\times$2 DiD in the data. Some of those 2$\times$2s use *already-treated* units as controls. When effects are dynamic, that's poison.
2. **Callaway & Sant'Anna (2021), the fix.** Forward-engineer from the target parameter ATT(g, t). Estimate each one with the never-treated or not-yet-treated — never the already-treated. Then aggregate.

The afternoon-morning thread: yesterday's DR-DiD is the engine inside CS's ATT(g,t).

LESSON

1. Bacon's decomposition

TWFE is a weighted average of 2×2 DiDs — and OLS picks the weights

Bacon's claim: TWFE *is* a weighted average of 2×2s

The TWFE regression with K timing groups produces an estimate $\widehat{\delta}^{\text{TWFE}}$ that is *numerically identical* to a weighted average of every 2×2 DiD you can construct from the data.

The decomposition (informally)

$$\widehat{\delta}^{\text{TWFE}} = \sum_{\text{every } 2\times 2 \text{ in the data}} s_{ij}\, \widehat{\delta}_{ij}^{2\times 2}$$

- This is *pure* OLS algebra. No assumptions about treatment effects yet.
- It's also not surprising: any OLS coefficient is a weighted average of some underlying "comparison" — Angrist (1998), Tymon (2022) for the general result.
- The interesting questions: **which 2×2s, with what weights, and is the average we get the average we want?**

With K timing groups, there are many 2×2s

Three timing groups $\{a, b, c\}$ plus one never-treated group U. Bacon counts the 2×2s:

Treated vs. Never	Treated vs. Later	Treated vs. Earlier
a vs. U	a vs. b (later)	b vs. a (earlier)
b vs. U	a vs. c (later)	c vs. a (earlier)
c vs. U	b vs. c (later)	c vs. b (earlier)

- Column 1: the comparisons you intended.
- Column 2: still uses an *eventually*-treated unit as a control — it's fine, because that unit hasn't been treated yet in the comparison window.
- Column 3: uses an *already*-treated unit as a control. **This is the forbidden comparison.**

Two timing groups: k (early) and ℓ (late)

Three time windows: PRE, MID (between the two treatment dates), POST.

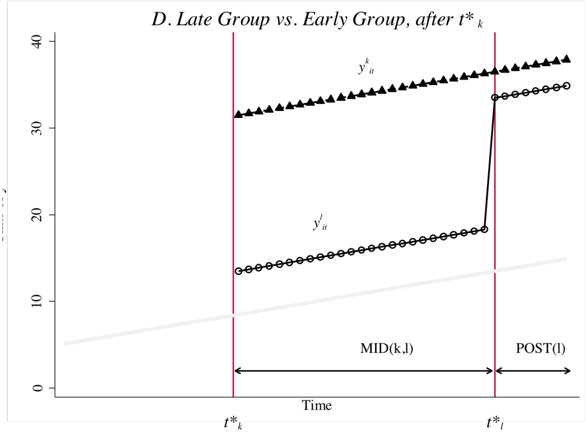
- **k vs. ℓ during MID.** Group k is post-treatment; group ℓ is still pre-treatment. **Comparison is fine** — ℓ is a not-yet-treated control.

$$\widehat{\delta}_{k\ell}^{2\times 2,\,k} = \left(\bar{y}_k^{\text{MID}} - \bar{y}_k^{\text{PRE}}\right) - \left(\bar{y}_\ell^{\text{MID}} - \bar{y}_\ell^{\text{PRE}}\right)$$

- **ℓ vs. k during POST.** Group ℓ is post-treatment; group k is *also* post-treatment. **The forbidden comparison** — k is now an already-treated control.

$$\widehat{\delta}_{\ell k}^{2\times 2,\,\ell} = \left(\bar{y}_\ell^{\text{POST}} - \bar{y}_\ell^{\text{MID}}\right) - \left(\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}}\right)$$

The forbidden 2×2, visualized



Group ℓ treated in Post*; group k already treated since* Mid*. One side is a treated unit that should never have been a control.*

The full Bacon decomposition

The setup. K timing groups, possibly one never-treated group U. TWFE estimate:

$$\widehat{\delta}^{\text{TWFE}} = \underbrace{\sum_{k \neq U} s_{kU}\, \widehat{\delta}^{2\times 2}_{kU}}_{\text{treated vs. never-treated}} + \underbrace{\sum_{k \neq U} \sum_{\ell > k} s_{k\ell} \Big[\mu_{k\ell}\, \widehat{\delta}^{2\times 2,\,k}_{k\ell} + (1 - \mu_{k\ell})\, \widehat{\delta}^{2\times 2,\,\ell}_{\ell k} \Big]}_{\text{treated vs. another-timing-group}}$$

- First sum: every timing group compared to the never-treated. Clean.
- Second sum: every *pair* of timing groups (k, ℓ) with k earlier, ℓ later, contributes two 2×2s — one with ℓ as not-yet-treated control (fine), one with k as already-treated control (**forbidden**).
- Weights s_{kU}, $s_{k\ell}$, and the within-pair weight $\mu_{k\ell}$ come from OLS algebra, not theory.

Notation: we write the forbidden 2×2 as $\widehat{\delta}^{2\times 2,\ell}_{\ell k}$. Bacon (2021, Eq. 10a) writes it as $\widehat{\beta}^{2\times 2,\ell}_{k\ell}$. Same object; we follow Scott's CI-2 convention.

Where the weights come from: variance of $\tilde{D}$

After de-meaning D_{it} within unit and time, what remains is $\tilde{D}_{it}$. The Bacon weights are functions of $\widehat{\mathrm{Var}}(\tilde{D}_{it})$ and group shares:

$$
\begin{aligned}
s_{kU} &= \frac{n_k\, n_U\, \overline{D}_k(1-\overline{D}_k)}{\widehat{\mathrm{Var}}(\tilde{D}_{it})} \\
s_{k\ell} &= \frac{n_k\, n_\ell\, (\overline{D}_k-\overline{D}_\ell)(1-(\overline{D}_k-\overline{D}_\ell))}{\widehat{\mathrm{Var}}(\tilde{D}_{it})} \\
\mu_{k\ell} &= \frac{1-\overline{D}_k}{1-(\overline{D}_k-\overline{D}_\ell)}
\end{aligned}
$$

- $\overline{D}_k$ is the share of group k's panel spent in treatment.
- Same variance-weighting Angrist (1998) and Tymon (2022) describe generally: OLS weights treatment contrasts by treatment-status variance.

Variance is maximized in the middle of the panel

Consider the never-treated 2×2 weight, s_{kU}. The key factor: $\overline{D}_k(1-\overline{D}_k)$.

$\overline{D}_k$ (share of panel treated)	$\overline{D}_k(1-\overline{D}_k)$
0.1	0.09
0.3	0.21
0.5	$\mathbf{0.25}$ $\leftarrow$ **max**
0.7	0.21
0.9	0.09

- Groups treated in the *middle* of the panel get the highest weight.
- Same logic for $(\overline{D}_k - \overline{D}_\ell)\big(1-(\overline{D}_k - \overline{D}_\ell)\big)$.
- **Panel length affects the weights.** Trim the panel and you've changed the estimand.

So far: pure algebra. Now: potential outcomes.

What we have:

- TWFE is a weighted average of 2×2s.
- Some of those 2×2s use already-treated units as controls.
- OLS picks the weights based on treatment variance.

What we don't yet have:

- Whether the weighted average we get *is* the weighted ATT we want.

To answer that we need to switch from realized outcomes to potential outcomes.

The next four slides walk through a single 2×2 step by step, then substitute back into the decomposition. The result is a clean expression for the bias.

LESSON

2. The forbidden 2×2, step by step

Add a zero, rearrange, and the bias appears

Step 1 — write down the 2×2 in observed outcomes

The 2×2 we're worried about: group ℓ (later-treated) in Post, with group k (earlier-treated) as control.

$$\boxed{\widehat{\delta}_{\ell k}^{2\times 2,\,\ell} = \left[\bar{y}_\ell^{\text{POST}} - \bar{y}_\ell^{\text{MID}}\right] - \left[\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}}\right]}$$

- Both groups are *treated* in Post.
- Both groups were *treated* in Mid? No — ℓ is still untreated in Mid.
- So group ℓ's "change" includes its move from untreated to treated. Group k's "change" is from *treated* to *treated*.

Step 2 — switch to potential outcomes

Replace each observed mean with the appropriate potential-outcome mean (using the switching equation):

$$\begin{aligned}
\bar{y}_\ell^{\text{POST}} &= \mathbb{E}[Y^1 \mid \ell, \textsc{Post}] && \text{(treated)} \\
\bar{y}_\ell^{\text{MID}} &= \mathbb{E}[Y^0 \mid \ell, \textsc{Mid}] && \text{(not yet treated)} \\
\bar{y}_k^{\text{POST}} &= \mathbb{E}[Y^1 \mid k, \textsc{Post}] && \text{(treated)} \\
\bar{y}_k^{\text{MID}} &= \mathbb{E}[Y^1 \mid k, \textsc{Mid}] && \text{(already treated)}
\end{aligned}$$

Notice the asymmetry. Group k is in Y^1 in *both* MID and POST. That's the problem.

Three of the four cells are Y^1. Only one is Y^0.

Step 3 — add zeros

We add three zero-valued terms of the form $\mathbb{E}[Y^0 \mid \cdot] - \mathbb{E}[Y^0 \mid \cdot]$:

The three zeros

1. $+\mathbb{E}[Y^0 \mid \ell, \textsc{Post}] - \mathbb{E}[Y^0 \mid \ell, \textsc{Post}]$ (add to ℓ in Post)
2. $+\mathbb{E}[Y^0 \mid k, \textsc{Post}] - \mathbb{E}[Y^0 \mid k, \textsc{Post}]$ (add to k in Post)
3. $+\mathbb{E}[Y^0 \mid k, \textsc{Mid}] - \mathbb{E}[Y^0 \mid k, \textsc{Mid}]$ (add to k in Mid)

- Each adds zero. The 2×2 is unchanged.
- But now we can group the terms into something interpretable.

Step 4 — rearrange into ATT, parallel-trends, and ΔATT

Group the eight terms (four observed + four added zeros) into three buckets:

The forbidden 2×2 in causal pieces

$$\widehat{\delta}_{\ell k}^{2\times 2,\,\ell} = \underbrace{\text{ATT}_\ell(\textsc{Post})}_{\text{what we want}} + \underbrace{\Delta Y_\ell^0(\textsc{Post},\textsc{Mid}) - \Delta Y_k^0(\textsc{Post},\textsc{Mid})}_{\text{parallel-trends bias}} - \underbrace{\left[\text{ATT}_k(\textsc{Post}) - \text{ATT}_k(\textsc{Mid})\right]}_{\text{heterogeneity bias: } \Delta\text{ATT}_k}$$

- Bucket 1 is the ATT we want for group ℓ in the post-period.
- Bucket 2 is the standard PT bias — requires PT between ℓ and k in this window.
- Bucket 3 is **new**: it's the *change* in group k's own ATT between Mid and Post. If treatment effects grow (dynamics), this is positive — and it enters with a *minus sign*.

Step 5 — substitute back into Bacon's decomposition

The three flavors of 2×2 in causal pieces:

$$\begin{aligned}
\widehat{\delta}^{2\times 2}_{kU} &= \text{ATT}_k(\textsc{Post}) \;+\; \Delta Y^0_k(\text{post},\text{pre}) - \Delta Y^0_U(\text{post},\text{pre}) \\
\widehat{\delta}^{2\times 2,\,k}_{k\ell} &= \text{ATT}_k(\textsc{Mid}) \;+\; \Delta Y^0_k(\textsc{Mid},\text{pre}) - \Delta Y^0_\ell(\textsc{Mid},\text{pre}) \\
\widehat{\delta}^{2\times 2,\,\ell}_{\ell k} &= \text{ATT}_\ell(\textsc{Post}) \;+\; \Delta Y^0_\ell(\textsc{Post},\textsc{Mid}) - \Delta Y^0_k(\textsc{Post},\textsc{Mid}) \\
&\quad - \;\Delta\text{ATT}_k
\end{aligned}$$

Plug these into Bacon's weighted sum. The TWFE probability limit:

$$\textbf{plim}\,\widehat{\delta}^{\textbf{TWFE}} = \textbf{VWATT} + \textbf{VWPT} - \boldsymbol{\Delta}\textbf{ATT}$$

Bacon (2021, Eq. 15) writes VWCT (variance-weighted common trends). Same object as our VWPT.

What the decomposition tells us

$$\operatorname{plim} \widehat{\delta}^{\mathsf{TWFE}} = \mathsf{VWATT} + \mathsf{VWPT} - \Delta\mathsf{ATT}$$

- **VWATT** — a variance-weighted ATT. Not the ATT *you* want unless you specifically wanted variance-weighted averaging. The weights depend on *panel length* (a researcher choice).
- **VWPT** — variance-weighted parallel-trends bias. Zero only if PT holds across *every* pairwise comparison, including between treated groups.
- **ΔATT** — the dynamics term. If ATT grows with time-since-treatment, this is *positive* and enters with a *minus sign* — attenuating TWFE toward zero, or even **flipping the sign**.

That's exactly what happens in baker.do: dynamics big enough that $-\Delta$ATT swamps the rest, flipping $+82$ to -6.69.

LESSON

3. baker.do, in code

Set the seed, write the DGP, compute the truth, run TWFE, watch it fail

The panel: 30 explicit lines, because repetition *is* the lesson

```
set obs 40
gen state = _n
expand 25
bysort state: gen firms = runiform(0,5)
expand 30
sort state
bysort state firms: gen year = _n

replace year = 1980 if year==1
replace year = 1981 if year==2
replace year = 1982 if year==3
replace year = 1983 if year==4
replace year = 1984 if year==5
replace year = 1985 if year==6
replace year = 1986 if year==7 // cohort 1 first treated here
replace year = 1987 if year==8
replace year = 1988 if year==9
... (22 more explicit lines, one per year, through 2009) ...
```

A loop would be shorter. The 30 lines are the lesson: each is one substitution, one beat. See `simulations/README.md` §1.

The DGP, in 12 more explicit lines

```
set seed 20200403

* Group-specific TE means (heterogeneous across cohorts)
gen te1 = rnormal(10, 0.04) // cohort 1986
gen te2 = rnormal( 8, 0.04) // cohort 1992
gen te3 = rnormal( 6, 0.04) // cohort 1998
gen te4 = rnormal( 4, 0.04) // cohort 2004
gen te = cond(group==1, te1, cond(group==2, te2, cond(group==3, te3, te4)))

* Y(0) and Y(1) under DYNAMIC effects (te grows since treatment)
gen y0 = firms + n + e
gen te_dynamic = te*treat*(year - treat_date + 1)
gen y = y0 + te_dynamic
```

One step per line, named variables, no loops. Full file: `simulations/staggered/baker.do`.

The truth, computed before running anything

```
* The truth (averaged across cohorts and post-periods)
su te_dynamic if treat == 1
display "True average ATT (dynamic) = " r(mean)
```

Output: `True average ATT (dynamic) = 82`

Why 82? The dynamic effect for group 1 averages $(10 + 20 + \cdots + 240)/24$. Combined across cohorts and post-periods, the simple average is 82.

We computed the truth from the DGP. Now compare it to what TWFE produces.

TWFE, applied

```
areg y i.year treat, a(id) robust
```

	Truth	TWFE
$\widehat{\text{ATT}}$	$+82$	-6.69^{***}

Sign-flipped. Extreme dynamics — the $-\Delta$ATT term swamps VWATT.

The Bacon decomposition of that -6.69

DD comparison	Weight	Avg DD estimate
Earlier-treated vs. later-treated-(as-control)	0.500	$+51.80$
Later-treated vs. earlier-treated-(as-control)	0.500	-65.18

$$(0.5)(+51.80) \; + \; (0.5)(-65.18) \; = \; -6.69.$$

Read it. Half the weight is on a 2×2 averaging -65 — a 2×2 that uses an already-treated group as control. **That's the forbidden comparison doing the damage.**

Lesson

4. Visualizing staggered structure

What the cells look like, what TWFE averages over

Constant-TE worksheet

Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1980–1985	0	0	0	0
1986–1991	10	0	0	0
1992–1997	10	8	0	0
1998–2003	10	8	6	0
2004–2009	10	8	6	4
ATT(g)	10	8	6	4

- **Simple ATT** (uniform avg over colored cells) $= 8.0$.
- **VWATT** (TWFE's target) $\neq 8.0$ — OLS weights down edge cohorts via $\overline{D}(1 - \overline{D})$.

Even with constant effects per cohort, TWFE doesn't give the simple ATT.

Dynamic-TE worksheet

Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1986	10	0	0	0
1992	70	8	0	0
1998	130	56	6	0
2004	190	104	42	4
2009	240	144	72	24

ATT grows each year since treatment (the dynamic DGP). Simple ATT (uniform avg over colored cells) ≈ 82; TWFE $= -6.69$.

The $-\Delta$ATT bias is doing the damage. Late cohorts use the already-treated earlier cohorts as "controls" — whose outcomes are growing.

LESSON

5. Callaway & Sant'Anna

Forward-engineer from ATT(g, t), never use the already-treated as control

The forward-engineering principle

Bacon told us what reverse-engineering from TWFE buys you: a variance-weighted average of every 2×2 in the data, including the forbidden ones.

The Callaway-Sant'Anna (2021) inversion:

1. **Start** with the target parameter you actually want: ATT(g,t) — the average treatment effect on cohort g in calendar period t.
2. **State** the identification assumptions formally.
3. **Derive** an estimator directly from those assumptions.
4. **Aggregate** the building blocks into whatever summary you want.

Each ATT(g,t) is its own clean 2×2 problem. Yesterday afternoon's DR-DiD is exactly the engine that estimates one.

ATT(g,t) — the building block

Definition

$$\text{ATT}(g,t) \;=\; \mathbb{E}\big[\, Y_{i,t}(1) - Y_{i,t}(0) \,\big|\, G_i = g \,\big]$$

- $G_i = g \;=\;$ "unit i was first treated in period g." (Cohort identifier.)
- One ATT(g,t) for every (cohort, calendar-period) pair where the cohort is treated.
- In baker.do: 4 cohorts $\times$ (varying numbers of post-periods) $=$ 60 ATT(g,t) cells in the grid.
- Each cell is a single 2$\times$2 estimand. **The aggregation is a separate, downstream choice.**

Parallel trends in a staggered setting — two choices of control

You can ask the parallel-trends assumption to hold against either of two control groups:

- **Never-treated units.** Cleanest comparison; needs at least one such group to exist.
- **Not-yet-treated units.** Late-treated cohorts serve as controls for earlier cohorts *during their own pre-treatment window*. Never as already-treated controls.

Either way, **the forbidden 2×2 of Bacon is ruled out by construction.** CS never uses an already-treated unit as a comparison.

If you have a clean never-treated group, use it. If you don't, use not-yet-treated. The math is the same.

The estimator — and yesterday's DR is the recommended one

Three options for the building-block estimator of each ATT(g, t):

1. **OR.** Heckman-Ichimura-Todd: model $\Delta Y(0)$ on X, impute the treated counterfactual.
2. **IPW.** Abadie (2005): weight controls by $p(X)/(1 - p(X))$.
3. **DR.** Sant'Anna-Zhao (2020). Consistent if *either* model is correct.

The practitioner's-guide recommendation: "Favor doubly robust."

$$\widehat{\delta}_{\text{DR}}(g, t) = \mathbb{E}\Big[w_{\text{DR}}(X) \left(\Delta Y - \widehat{\mu}_0(X)\right) \Big]$$

The DR-DiD you proved yesterday afternoon. Same engine, applied cell-by-cell to every feasible (g, t).

Aggregation is a downstream choice, not an estimation choice

Once you have $\{\widehat{\text{ATT}}(g,t)\}$, build whatever summary the policy question demands:

- **Simple ATT.** Uniform average over all post-treatment (g,t) cells.
- **Group ATT, ATT(g).** Average each cohort's own post-treatment cells. "What did the policy do to cohort g on average?"
- **Event-time ATT, ATT$_{\text{es}}(e)$.** Average across cohorts at fixed event time e since treatment. "What's the dynamic profile?"
- **Calendar-time ATT.** Average across cohorts within a calendar period. "What happened in 2002, on average?"

Inference. Bootstrap, with uniform confidence bands when reporting an event-study (the `did` R package does this by default).

baker_cs.do — the implementation

```
* Get the same baker data, fit CS with DR-DiD
ssc install csdid, replace
ssc install drdid, replace

csdid y, ivar(id) time(year) gvar(treat_date) ipw long2

* Four aggregations, four answers:
csdid_estat simple // simple ATT across all cells
csdid_estat group // ATT(g) per cohort
csdid_estat calendar // ATT by calendar period
csdid_estat event // event-time profile
```

Building blocks first, aggregation second.

collegio-carlo-alberto/labs/Baker/Solutions/baker_cs.do.

The headline: CS recovers the truth

	Truth	TWFE	CS (DR-DiD)
Feasible ATT (pre-2004 sample)	68.33	26.81^{***}	$\mathbf{68.34}^{***}$

- The “feasible” sample drops the 2004 cohort so a never-treated comparison exists.
- TWFE is no longer sign-flipped (no universally-treated tail), but it still attenuates — still has VWATT weights, still uses early-treated cohorts as controls.
- CS lands on the truth to two decimals.

Not every ATT(g, t) is feasible

CS can only estimate ATT(g, t) if there is a valid comparison group for (g, t):

- **Never-treated comparison:** you need at least one unit untreated at t.
- **Not-yet-treated comparison:** you need at least one unit that hasn't *yet* reached its treatment date by t.

In baker.do: The last cohort is treated in 2004. After 2004, every unit is treated. **No comparison units exist for ATT(g, t > 2004).**

Two responses:

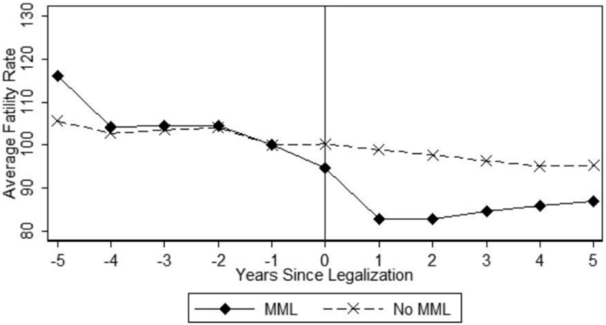
- Restrict to the feasible sample. CS will tell you which (g, t) it dropped.
- Or, if you really want estimates for the universally-treated tail, use a different framework (Imputation: Borusyak et al. 2024; Wooldridge 2021) that doesn't require a comparison group.

Lesson

6. Sun & Abraham (2021)

Bacon was static. Event-studies have their own contamination.

Researchers love event studies — and TWFE will lie to you on them



Anderson et al. (2013): raw traffic-fatality rates for treated states (re-centered to treatment date) vs. controls. The classic dynamic-effects display.

The classic dynamic-TWFE event-study spec

Researchers want to plot pre-treatment placebo coefficients and post-treatment dynamic effects. The standard regression:

$$Y_{i,t} \;=\; \alpha_i \;+\; \delta_t \;+\; \sum_{g \in G} \mu_g \, \mathbf{1}\{t - E_i \in g\} \;+\; \varepsilon_{i,t}$$

- E_i is unit i's treatment date. $t - E_i$ is event time (relative-to-treatment).
- G is the set of event-time bins (leads and lags) included.
- Coefficient μ_g is supposed to be the average treatment effect at event-time bin g.

Bacon told us static TWFE is biased. Sun & Abraham (2021) ask: *what about the dynamic version?*

Each lead and lag picks up treatment effects from *other* leads and lags

Write $\text{ATT}_{e,l}$ for the treatment effect on cohort e at l years from its treatment date.

Even under parallel trends and no anticipation, the TWFE coefficient on event-time bin g is:

$$\mu_g = \underbrace{\textstyle\sum_{l\in g}\sum_e w^g_{e,l}\,\text{ATT}_{e,l}}_{\text{the target (own bin)}}$$

$$+ \underbrace{\textstyle\sum_{g'\neq g}\sum_{l'\in g'}\sum_e w^g_{e,l'}\,\text{ATT}_{e,l'}}_{\text{leakage from \textit{other} included bins}}$$

$$+ \underbrace{\textstyle\sum_{l'\in g^{\text{excl}}}\sum_e w^g_{e,l'}\,\text{ATT}_{e,l'}}_{\text{leakage from \textit{excluded} bins}}$$

The cross-bin weights sum to zero — but only cancel when treatment effects are identical across cohorts. Once they're heterogeneous, the leakage doesn't cancel.

The disturbing corollary: pre-period μ_g can be nonzero even when PT holds

Under PT $+$ no anticipation but *not* homogeneity:

- A pre-period lead μ_{-2} should be zero if pre-trends are truly parallel.
- In TWFE it isn't — it picks up post-treatment effects from *other* cohorts through the cross-bin weights.
- So a “failed placebo” may just be the cross-bin leakage.
- And a “passing placebo” may just be cancellation, not evidence of parallel trends.

Lesson: Don't drop excluded bins arbitrarily. Lagged effects contaminate through the excluded bins.

Same disease as Bacon's static decomposition, but applied to leads and lags — so it affects the entire plot you're showing the audience.

SA's fix: the interaction-weighted estimator (3 steps)

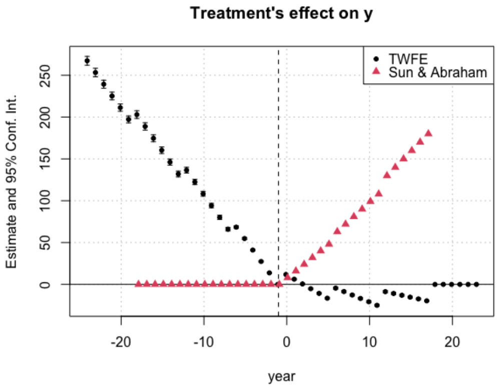
1. **Estimate each $\widehat{\delta}_{e,l}$ separately.** A saturated regression with cohort $\times$ event-time interactions, using the never-treated (or last-treated) as control:

$$Y_{i,t} \;=\; \alpha_i \;+\; \lambda_t \;+\; \sum_{e \notin C} \sum_{l \neq -1} \delta_{e,l} \left[\mathbf{1}\{E_i = e\} \cdot D_{i,t}^{l} \right] \;+\; \varepsilon_{i,t}$$

2. **Estimate cohort shares** in the relevant periods: $\widehat{\Pr}(E_i = e \mid E_i \in [-l, T-l])$.
3. **Aggregate** the $\widehat{\delta}_{e,l}$ using the cohort shares as weights.

Same logic as CS. Estimate cohort-event-time cells first; aggregate after.
Forward-engineer the parameter. *Software: `eventstudyinteract` (Stata, SSC); `fixest::sunab` (R).*

TWFE event study vs. SA, side by side



TWFE leads slope downward pre-treatment — the cross-bin contamination. SA's pre-period sits at zero; lags trace the dynamic profile.

Lesson

7. de Chaisemartin & D'Haultfœuille

Negative weights on *unit-level* effects, plus the on-off case

dCdH (2020) adds three things to the staggered toolkit

1. **A different decomposition.** Bacon decomposed TWFE into 2×2 building blocks with *positive* weights. dCdH decomposes TWFE into *unit-level* treatment effects — where weights can be **negative**.
2. **The on-off case.** CS and SA assume irreversible treatment. dCdH handles units that switch into and out of treatment (think: a state policy that was repealed). Roe v. Wade adopted in 1973, overturned in 2022.
3. **A diagnostic.** Run the TWFE you were going to run, then count the *share of unit-period treatment effects* that receive negative weights. Report the sign-flip threshold.

The dCdH decomposition: weights on *cell-level* treatment effects

Let $\Delta_{g,t}$ be the average treatment effect for group g in period t — one number per treated cell of the panel. Under parallel trends and no anticipation:

Theorem 1 (dCdH 2020)

$$\beta_{\text{fe}} = \mathbb{E}\left[\sum_{(g,t):\,D_{g,t}=1} \frac{N_{g,t}}{N_1}\, w_{g,t}\,\Delta_{g,t}\right], \quad \sum_{(g,t)} \frac{N_{g,t}}{N_1} w_{g,t} = 1, \quad w_{g,t} \textbf{ can be negative}.$$

- The weights $w_{g,t}$ are built from the residuals of regressing $D_{g,t}$ on group and time fixed effects.
- Bacon decomposes the *estimator* into 2×2 building blocks (always positive). dCdH decompose the *estimand* into cell-level effects (**may be negative**).
- And dCdH applies to *any* TWFE design, not just staggered.

Negative weights break the "no sign-flip property"

Thought experiment. Suppose every treated cell gained from the treatment ($\Delta_{g,t} > 0$ for all (g, t)).

- If all $w_{g,t} > 0$: TWFE is a positive weighted average of positive effects. Sign matches.
- If some $w_{g,t} < 0$: nothing prevents TWFE from being **negative** — even though every cell gained.

This is what we saw in baker.do. Truth $+82$, TWFE -6.69. Some cells got large negative weights; they swamped the positive-weight contributions. *Pre-dCdH, "OLS is a weighted average" was reassuring. dCdH made it disturbing.*

When the negative weights bite (and when they don't)

Run two diagnostics on your data before publishing:

1. **The share of treated cells with negative weight.** If a meaningful fraction of cells gets a minus sign, your headline is built on those cancellations.
2. **The smallest amount of heterogeneity in $\Delta_{g,t}$ that could flip the sign of TWFE.** If small, your sign is fragile. If large, your sign is robust.

Rules of thumb:

- A big never-treated group helps: fewer negative weights, smaller magnitudes.
- Treatment dates packed close together: also fewer.
- A long panel with a few early-treated cohorts: the worst case.

Software: `twowayfeweights` (Stata SSC); R via `TwoWayFEWeights`. Both numbers come out on your specific dataset.

The on-off case: `did_multiplegt` for switching treatment

Suppose treatment can switch on *and* off (Roe v. Wade 1973 $\rightarrow$ Dobbs 2022; minimum-wage rollbacks; tax cuts that sunset).

- CS and SA both assume **irreversible** treatment — they cannot estimate effects in this setting.
- dCdH propose a different estimator built on *instantaneous* effects between adjacent periods, comparing units that just switched against units whose treatment status is unchanged.
- Equivalent to CS & SA when treatment is irreversible (with their particular weighting), but *also* handles the on-off case.

Software: `did_multiplegt` (Stata); the `DIDmultiplegt` R package.

The unifying lesson across Bacon, SA, and dCdH

What makes weights "good"

The economic question is: *what parameter do you want?*

OLS doesn't ask. OLS picks weights from the residual variance of D.

- Bacon: TWFE = weighted average of $2{\times}2$s. Weights positive, driven by treatment-status variance.
- SA: μ_g = weighted average of treatment effects across cohorts and event-times. Cross-bin leakage.
- dCdH: TWFE = weighted average of cell-level treatment effects $\Delta_{g,t}$. **Weights can be negative.**

All three say the same thing: if you let OLS pick the weights, you've let it pick your economic question. Forward-engineer instead.

Lesson

8. Borusyak-Jaravel-Spiess imputation

The Heckman-Ichimura-Todd move, lifted to fixed effects

All causal inference is imputation

Imbens & Rubin (2015), *Causal Inference*

"At some level, all methods for causal inference can be viewed as imputation methods, although some more explicitly than others."

What's the missing thing? Always Y^0 for the treated. What we never observe.

What's the estimator? A model for Y^0 fit on the people who reveal it (the untreated), then plugged in for the treated.

You already saw one today. Yesterday afternoon's OR-DiD was imputation. BJS is the same trick, with a different model.

Yesterday's HIT / OR-DiD recipe (recap)

Heckman, Ichimura, Todd (1997). The conditional-PT version of imputation:

1. **Fit the model** of the untreated trend on X, using only $D = 0$:

$$\widehat{\mu}_0(X) \;=\; \widehat{\mathbb{E}}[\Delta Y \mid D{=}0,\; X]$$

2. **Impute** each treated unit's counterfactual trend $\widehat{\mu}_0(X_i)$.
3. **Average** the residual on the treated:

$$\widehat{\delta}_{\text{OR}} \;=\; \frac{1}{N_1} \sum_{i:D_i=1} \big(\Delta Y_i - \widehat{\mu}_0(X_i)\big)$$

The counterfactual model is on ΔY, indexed by X.

The BJS (2024) recipe, side-by-side

Borusyak, Jaravel, Spiess (2024). Same imputation logic, but the counterfactual model is on *levels* $Y(0)$ indexed by *fixed effects*:

1. **Fit the model** of untreated $Y(0)$ on unit FE $+$ time FE (and any pre-treatment X), using only $D = 0$ observations:

$$Y(0)_{it} \;=\; \alpha_i \;+\; \lambda_t \;+\; X_{it}'\delta \;+\; \varepsilon_{it}$$

2. **Impute** the counterfactual for each treated (i,t): $\widehat{Y}_{it}^{0} \;=\; \widehat{\alpha}_i + \widehat{\lambda}_t + X_{it}'\widehat{\delta}$.
3. **Average** the residual on the treated unit-periods:

$$\widehat{\tau}_{it} \;=\; Y_{it}^{1} - \widehat{Y}_{it}^{0}, \qquad \widehat{\delta}_W \;=\; \sum_{(i,t)\in\Omega_1} w_{it}\,\widehat{\tau}_{it}$$

Same three moves as HIT. Only the counterfactual model differs.

Why BJS avoids the forbidden contrast

The fixed-effects regression in Step 1 is fit on **untreated observations only.**

- No treated observation enters the model that predicts $Y(0)$.
- Already-treated units contribute their pre-treatment periods only (those are still untreated).
- The fixed effects propagate across time, so $\widehat{\alpha}_i$ for unit i can still be estimated even if i is eventually treated — using i's own pre-treatment observations.

Compare to TWFE. TWFE fits δ *and* the fixed effects on *all* observations simultaneously — so a treated unit's post-period influences the estimated FE, which feeds back into the residual that defines $\widehat{\delta}$. That's the forbidden-contrast channel.

BJS just refuses to let that happen. Same model class, different fitting sample.

BJS compared with TWFE, CS, SA on the same DGP

Table 3: Efficiency and Bias of Alternative Estimators

Horizon	Estimator	Baseline simulation		More pre-periods	Heterosk. residuals	AR(1) residuals	Anticipation effects
		Variance (1)	Coverage (2)	Variance (3)	Variance (4)	Variance (5)	Bias (6)
$h = 0$	Imputation	0.0099	0.942	0.0080	0.0347	0.0072	-0.0569
	DCDH	0.0140	0.938	0.0140	0.0526	0.0070	-0.0915
	SA	0.0115	0.938	0.0115	0.0404	0.0066	-0.0753
$h = 1$	Imputation	0.0145	0.936	0.0111	0.0532	0.0143	-0.0719
	DCDH	0.0185	0.948	0.0185	0.0703	0.0151	-0.0972
	SA	0.0177	0.948	0.0177	0.0643	0.0165	-0.0812
$h = 2$	Imputation	0.0222	0.956	0.0161	0.0813	0.0240	-0.0886
	DCDH	0.0262	0.958	0.0262	0.0952	0.0257	-0.1020
	SA	0.0317	0.950	0.0317	0.1108	0.0341	-0.0850
$h = 3$	Imputation	0.0366	0.928	0.0255	0.1379	0.0394	-0.1101
	DCDH	0.0422	0.930	0.0422	0.1488	0.0446	-0.1087
	SA	0.0479	0.952	0.0479	0.1659	0.0543	-0.0932
$h = 4$	Imputation	0.0800	0.942	0.0546	0.3197	0.0773	-0.1487
	DCDH	0.0932	0.950	0.0932	0.3263	0.0903	-0.1265
	SA	0.0932	0.954	0.0932	0.3263	0.0903	-0.1265

Notes: See Section 4.6 for a detailed description of the data-generating processes and reported statistics.

TWFE event-study: biased pre-period leads, attenuated lags. CS, SA, BJS all recover the dynamic profile. Three different counterfactual models, same answer.

When BJS, when CS, when SA, when dCdH?

- **CS (DR).** You want ATT(g, t) building blocks. You have covariates that matter for conditional PT. **First choice when standard staggered.**
- **SA.** You specifically want an event-study plot. Equivalent to CS with never-treated; gives closed-form SEs.
- **BJS.** You want a single $\widehat{\tau}_{it}$ for every treated unit-period — maximum flexibility, can compute any aggregate. Comfortable assuming the parametric Y(0) model.
- **dCdH.** Your treatment switches *on and off*. CS, SA, BJS all assume irreversibility; dCdH does not.

Practitioner's guide framing: all four are forward-engineered estimators. Pick the one that matches the parameter you want and the assumptions you can defend.

LESSON

9. Honest DiD

You can't test parallel trends. You can bound the violation.

Parallel trends is not testable. Pretests are not the same assumption.

The standard ritual. Show that pre-period coefficients are statistically indistinguishable from zero. Conclude: "parallel trends holds."

Roth (2022) showed this is a mistake.

- Pre-trends measure differences in the *wrong* periods — they test whether trends were parallel *before* treatment, not whether they'd be parallel *after* in its absence.
- Pre-tests are typically **low-powered**: failing to reject "$\beta_{\text{pre}} = 0$" is consistent with non-trivial violation.
- Conditioning on the pretest passing *distorts* the post-period estimator (selection on the pretest outcome).

You cannot test PT. What you can do: bound the plausible violation.

Rambachan & Roth (2023): bound the violation, get partial-ID bounds

The reframe. Take the post-period PT violation
$\delta_t = \mathbb{E}[Y_t^0 - Y_{t-1}^0 \mid D{=}1] - \mathbb{E}[Y_t^0 - Y_{t-1}^0 \mid D{=}0]$ as an unknown nuisance.

1. Pick a *class* of plausible δ_t paths.
2. For each path in that class, the ATT is point-identified.
3. The union over the class gives a **partial-identification interval** for the ATT.

The output isn't a point estimate. It's a *robust confidence set*: "if the violation belongs to this class, the ATT lies in this interval."

Software: `HonestDiD` in R; `honestdid` in Stata (M. Cáceres Bravo). Plugs into the output of CS, SA, or BJS.

Two restriction classes: smoothness (M) and relative magnitudes ($\bar{M}$)

Smoothness (M-class). The violation can't accelerate too fast:

$$|\delta_{t+1} - \delta_t| \;\le\; M \quad \text{for all } t.$$

"Trajectory of any unobserved violation is at most M units off a line." Larger $M \Rightarrow$ more flexibility $\Rightarrow$ wider bounds. $M = 0$ pins down linear extrapolation.

Relative magnitudes ($\bar{M}$-class). Post-period violation $\le \bar{M}$ times the largest pre-period observed difference:

$$|\delta_{\text{post}}| \;\le\; \bar{M} \cdot \max_{\text{pre}} |\widehat{\delta}_{\text{pre}}|.$$

"If the pre-trend looks small, the post-trend can't be much bigger." Common values $\bar{M} \in \{0.5,\, 1,\, 2\}$.

Pick the class that matches what you'd defend at the seminar.

The breakdown value: how much PT violation would erase your result?

For each M (or $\bar{M}$), Rambachan-Roth give you a confidence set $\mathcal{C}(M)$ for the ATT.

Breakdown value

$$M^{\text{break}} = \min\{\, M \;:\; 0 \in \mathcal{C}(M) \,\}.$$

Read it. "If parallel trends is allowed to deviate by more than M^{break} per period, my significant result could vanish."

- Large breakdown value $\Rightarrow$ result is robust to plausible violations.
- Small breakdown value $\Rightarrow$ result is fragile; even tiny violations break significance.
- Reported *alongside* the headline, not in place of it.

This is the diagnostic you'll compute on the Brazil mental-health-reform application in the lab.

LESSON

10. Synthesis

What’s the same, what’s different, what to do tomorrow morning

Yesterday afternoon plus this morning $=$ one workflow

- **Yesterday (covariates).** Conditional PT $\Rightarrow$ DR-DiD. Two chances to be right (OR model or pscore model).
- **This morning (staggered).** ATT(g, t) building blocks, estimated cell-by-cell, then aggregated.
- **The combination.** Each ATT(g, t) *is* a 2$\times$2 problem. The recommended way to estimate it is DR-DiD with whatever X you need for conditional PT. Then aggregate.

One sentence

Forward-engineer the target, estimate the cells with DR, aggregate with non-negative weights.

When to use TWFE in staggered settings

Basically never.

- TWFE = ATT only when (a) treatment effects are homogeneous across cohorts, AND (b) treatment effects don't change over time within a cohort.
- Both fail in almost every applied setting.
- Even when they hold, the variance weights are still strange (depend on panel length).
- **If you must report it, also report CS, SA, BJS, or dCdH — and the dCdH negative-weight diagnostic. Explain the differences.**

Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025): "Do not use TWFE."

A student asks: "But what about my paper from last year that used TWFE?"

A reasonable thing to do, in order:

1. **Run the Bacon decomposition** (`ddtiming`; `bacondecomp` in R). See whether the forbidden 2×2s carry weight.
2. **Run the dCdH diagnostic** (`twowayfeweights`). Sizeable negative-weight share $\Rightarrow$ headline is suspect.
3. **Re-estimate with CS, SA, or BJS.** Compare the new headline to TWFE.
4. **If they disagree:** the difference *is* the $-\Delta$ATT / VWPT / negative-weight contamination. Trust the design-based estimator.
5. **If they agree:** in your setting, dynamics and heterogeneity didn't bite. Report both, explain.

Goodman-Bacon, Sant'Anna, Whitten (2026) gives a refined re-analysis framework for pre-trends-parallel settings.

Lab: the 9-step checklist on Dias & Fontes (2024)

The paper. CAPS mental-health centers rolling out across Brazilian municipalities, 2002–2016. Outcome: homicide rate.

1. Decide on the target parameter (write down what you want, then defend it).
2. Cohort count table. Identify never-treated, always-treated. Drop the 2002 cohort and say why.
3. Plot the treatment rollout (calendar grid in grayscale).
4. Plot the raw outcome by cohort. What do you see before estimation?
5. Covariate balance table (normalized differences, 2002 baseline).
6. Run TWFE first as a *diagnostic* (don't interpret causally yet).
7. Bacon decomposition & dCdH negative-weight diagnostic.
8. CS with DR-DiD: simple, group, calendar, event aggregations.
9. Honest DiD: $\bar{M}$-bounds and breakdown value.

Walkthrough: glasgow/labs/Brazil/README.md. Reference code: labs/Brazil/code/brazil.do and brazil.R. Data on Scott's GitHub: mixtape/brazil.dta.

Tomorrow afternoon

- **Honest event-studies.** Roth (2022) on conditioning on pretest passes. Rambachan & Roth (2023) partial-identification bounds.
- **Continuous and multi-valued treatment.** Dose-specific ATT, average causal response (Callaway, Goodman-Bacon, Sant'Anna 2024).
- **Goodman-Bacon, Sant'Anna, Whitten (2026)** — a refined re-analysis framework for pre-trends-parallel settings.
- **Two-stage DiD** (Gardner 2021) — the GMM cousin of BJS, with the same forward-engineered architecture.

Reading for tonight: *Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025)*
"Difference-in-Differences Designs: A Practitioner's Guide."

Forward-engineer.

Start with what you want to estimate.

Match the estimator to the assumption.
Aggregate at the end, not the start.

Scott Cunningham · CUNEF Auditorium · May 25–28, 2026